

SaVaGe User Manual

Product: SaVaGe — desktop image→SVG converter and vector editor

Platform: Windows-first (Tauri 2)

Document version: matches application v0.1.0

How to use this manual

If you are...	Start here
New to vector graphics	Quick start → Convert → Edit basics
Comfortable with Illustrator / Inkscape	Workspace map → Tools reference → Advanced
Integrating or extending SaVaGe	File formats → Plugins → Limitations

In the app: **Help** → **User Manual (PDF)**... opens this guide in your system PDF viewer.

Keyboard shortcuts appear in **bold** (for example **V** for Select). Menu paths use **File** → **Open...** style.

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1. Quick start

1.1 Launch

Run SaVaGe from your project folder (developer build):

```
$env:Path = "$env:USERPROFILE\.cargo\bin;$env:Path"
pnpm tauri:dev
```

Or open the installed desktop app if you built a release with `pnpm tauri:build`.

SaVaGe opens in **Convert** mode: dark graphite UI, lime accent, SaVaGe wordmark.

1.2 Your first conversion (beginner path)

1. Drag a PNG, JPEG, WEBP, GIF, BMP, or TIFF onto the drop zone — or click **Open Image...**
2. Pick a preset that matches the art:
 - **Logo / flat** — logos, icons, flat color art
 - **Photo** — photographs and soft gradients
 - **Line art** — sketches, comics, high-contrast line drawings
 - **Pixel** — pixel art / hard block shapes
3. Optionally tweak **Color precision**, **Filter speckle**, **Corner threshold**, **Path precision**, and **Mode**.
4. Click **Convert to SVG**.
5. When tracing finishes, SaVaGe switches to **Edit** mode with vector layers ready to refine.

1.3 Your first edit

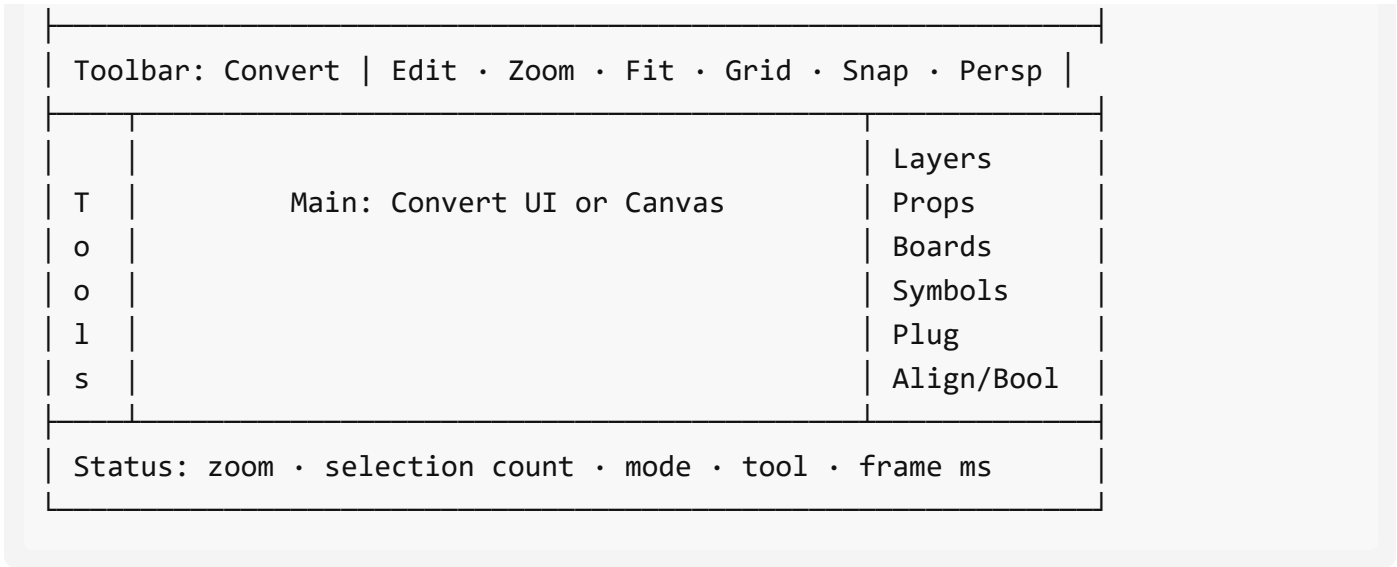
1. Press **V** (Select). Click a shape; drag to move. Use corner/edge handles to resize; use the rotate handle to rotate.
2. Open the **Props** tab on the right. Change fill color or opacity.
3. **File** → **Save Project...** and save a `.savage` file (best fidelity).
Or **File** → **Export SVG...** / **Export PNG...** for delivery.

1.4 Five-minute power path (experienced users)

Convert with **Logo** preset → Edit → boolean Unite overlapping paths → Shape Builder (**S**) to carve → Create Symbol → Mesh fill on a hero shape → Export SVG + keep `.savage` master.

2. Workspace map

```
| TitleBar: SaVaGe • File Edit Object View Help |
```



2.1 Modes

Mode	Purpose
Convert	Import raster art and trace it to SVG with vtracer (Rust)
Edit	Full vector scene: draw, style, boolean, symbols, export

Switch with the **Convert | Edit** control in the toolbar. Successful conversion also jumps to Edit automatically.

2.2 Status bar

Bottom strip shows:

- Zoom percentage
- Number of selected objects
- Current mode and active tool
- Last frame time in milliseconds (warns visually when over ~16 ms)

Transient toasts (boolean results, clip masks, plugin messages) appear over the main view.

3. Convert mode

Convert turns bitmaps into editable vector paths. Tracing runs in Rust (`vtracer`) for speed and quality.

3.1 Loading an image

Method	Notes
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Drag and drop	Preferred; uses the file path from the OS drop
Open Image...	File dialog for png / jpg / jpeg / gif / webp / bmp / tif / tiff

While converting, the drop zone is disabled and a progress indicator appears.

3.2 Presets (start here)

Presets apply a full option set. Choose the closest match, then fine-tune.

Preset	Best for	Technical tendency
Logo / flat	Logos, icons, UI graphics	Color mode; moderate color precision; light speck filtering; spline paths
Photo	Photos, soft shading	Slightly coarser color, stronger speck filter
Line art	Ink drawings, comics	Binary-style color treatment; very low color precision
Pixel	Pixel art	Pixel path mode; path precision 0; no speck filter

3.3 Trace options (detail)

Control	What it does	Amateur tip	Pro tip
Color precision	How finely colors are quantized	Higher = more colors / heavier SVG	Lower for flat brand marks; raise for photos
Filter speckle	Removes tiny islands	Increase if noisy	Trade detail vs cleanliness before Simplify Path
Corner threshold	Corner vs curve bias	Default is fine for logos	Lower for sharper corners; higher for smoother bends
Path precision	Numeric / geometric fidelity of paths	Leave default unless SVG is huge	Lower to reduce node count after convert
Mode	spline / polygon / pixel	Use spline for logos	polygon for faceted look; pixel for 1:1 pixel blocks

Some engine knobs (splice threshold, hierarchical clustering, layer difference, max dimension) are applied by presets internally and are not all exposed in the form.

3.4 Preview and continue

After conversion you get a side-by-side **Raster | SVG** preview when available. SaVaGe loads the SVG into the document store and opens **Edit**.

Open in Editor is available if SVG markup was already produced and you stayed on Convert.

3.5 Convert workflow tips

- Start with the smallest useful image that still has clean edges (very large photos produce heavy path sets).
 - Logo art with solid fills converts cleaner than anti-aliased soft shadows.
 - After convert, use **Object** → **Simplify Path** on noisy selections before booleans.
-

4. Edit mode basics

4.1 Navigating the canvas

Action	How
Pan	H (Pan tool), hold Space + drag, or middle-mouse drag
Zoom	Scroll wheel (zooms toward cursor)
Zoom buttons	Toolbar - / % / +
Fit active artboard	Toolbar Fit, View → Fit Artboard , or Ctrl+0
Fit selection	View → Fit Selection or Ctrl+2
Zoom 100%	Click zoom % , View → Zoom 100% , or Ctrl+1

4.2 Selecting and transforming

1. Activate **Select (V)**.
2. Click an object. **Shift+click** to add/remove from the selection.
3. Drag the body to move.
4. Drag handles to resize (**Shift** constrains).
5. Use the rotate handle to rotate.
6. Arrow keys nudge 1 px; **Shift+arrows** nudge 10 px.

Ctrl+A selects all. **Delete** / **Backspace** deletes the selection (ignored while typing in fields).

4.3 Undo and history

Action	Shortcut / menu
Undo	Ctrl+Z · Edit → Undo
Redo	Ctrl+Shift+Z or Ctrl+Y · Edit → Redo

History is document-scoped (Zundo). Prefer finishing a stroke before undoing mid-drag when possible.

4.4 Layers at a glance

Open the **Layers** tab:

- Top of the list $\approx$ front of the stack
- Click to select
- Toggle visibility and lock
- Double-click to rename
- Drag to reorder among root items
- Groups show nested children indented

4.5 Properties at a glance

Open **Props** (first selected object):

- Position **X / Y**, rotation **R**, opacity
 - **W / H** are informational (resize on-canvas with handles)
 - Fill and stroke paint editors
 - Stroke width
 - Effects: blur, drop shadow
 - Text and symbol-specific controls when relevant
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5. Tools reference

Tools live in the left rail (Edit mode only). Click an icon or use a shortcut.

5.1 Navigation & selection

Select — **V**

Hit-tests objects under the pointer. Supports move, multi-select (**Shift**), scale, and rotate via chrome handles.

Direct Select — **A**

Edits individual path points (anchors). Drag a point to move it (handles move with the point when present).

Variable width: hold **Alt** and drag a point horizontally to change that point's local stroke width. See [Variable-width strokes](#).

Pan — **H**

Drag the canvas. Prefer Space-pan for temporary panning without leaving your draw tool.

5.2 Shape tools

Tool	Shortcut	Draw	Modifiers
Rectangle	R	Drag	Shift square · Alt from center
Ellipse	O	Drag	Shift circle · Alt from center
Line	L	Drag	—
Polygon	—	Drag radius	Regular hexagon (6 sides)
Star	—	Drag radius	5-point star

Shapes land as scene nodes with default fill/stroke (brand lime accent where applicable).

5.3 Path tools

Pen — **P**

Click to place corner anchors. Drag while placing to create Bézier handles. Click near the first point to close.

Key	Action
Enter or Esc	Finish the path
Backspace	Remove the last point

Pencil

Freehand stroke simplified into an open path. Good for roughing silhouettes; refine with Direct Select or Simplify Path.

Brush (calligraphy) — **B**

Speed-sensitive calligraphy: faster motion → thinner ribbon. Result is a **closed filled path** (not a uniform stroke), which booleans and fills treat as solid geometry.

5.4 Pattern Brush & Scatter Brush

Tool	Result
Pattern Brush	Chevron motifs stamped along your stroke at regular spacing, tangent-aligned, grouped
Scatter Brush	Leaf-like motifs with position jitter, rotation noise, and scale variation, grouped

How to use: select the tool → drag a gesture → release. SaVaGe creates a group containing stamp paths.

Pro note: stamps are real vector paths. Ungroup (or expand by detaching structure manually) if you need per-stamp boolean work. Spacing/jitter are currently fixed in-tool (not yet exposed as brush settings UI).

5.5 Shape Builder — S

Interactive region sculpting for overlapping filled shapes (rect, ellipse, closed path; up to **four** selected shapes).

1. Select two or more overlapping shapes.
2. Activate **Shape Builder (S)**. SaVaGe decomposes atomic regions.
3. Regions tint **green** (kept) or **red** (discarded). Click a region to toggle.
4. Press **Enter** (or **Object** → **Commit Shape Builder** / sidebar button) to union kept regions into one path and remove sources.
5. **Esc** cancels and returns to Select.

Requires flattenable closed geometry. Open strokes and text are not valid region sources until converted/outlined.

5.6 Text — T

Click the canvas to place a text object (default "Text"), then edit inline. Double-click existing text to edit.

Enter commits; **Esc** cancels the overlay edit.

Typography fields: content, font size, weight (Props). Face defaults to the UI family (**DM Sans Variable**).

Convert Text to Outlines: Object menu or Props button. Implementation rasterizes the text glyph silhouette and extracts contours — useful for logo lockups, not a full font-outline engine. Prefer outlining late in the pipeline.

6. Panels

6.1 Layers

Hierarchical list of the scene graph. Use it for selection, visibility, locking, naming, and root reordering. Nested group children appear indented.

6.2 Props (Properties)

Context-sensitive inspector for the **first** selected node.

Paint editor (fill / stroke)

Type	Behavior
Solid	Single color
Linear	Linear gradient with editable stop colors
Radial	Radial gradient with editable stop colors
Mesh	Grid mesh gradient; edit corner (control point) colors
None	No paint

Stroke width uses a slider. Mesh fills are rasterized for canvas preview; SVG export approximates mesh with a patterned set of flat polygons (see limitations).

Path: variable width

For path objects:

- **Taper ends** — applies a sine-like width profile along each subpath
- **Clear profile** — removes per-point widths (returns to uniform stroke width)

Effects

- **Blur** — Gaussian-style blur via canvas filter / SVG filter on export
- **Drop shadow** — offset, blur, color/opacity (enable via shadow controls)

Clip

If the node has a clip mask, release it from Props or Object menu. Otherwise a short tip explains the make-mask workflow.

6.3 Boards (Artboards)

Multiple artboards live in one document.

Action	How
Activate	Click a board (fits view to it)
Add	+ or Object → New Artboard
Rename	Double-click the name
Delete	× (at least one board must remain)

The active artboard draws with a stronger accent frame and label on the canvas. View → **Fit All Artboards** frames every board.

6.4 Symbols

Reusable masters stored outside the main drawing tree; instances reference a master.

Action	How
Create	Select artwork → + / Create from selection / Object → Create Symbol
Place	Click a symbol name in the panel
Detach	Select an instance → Detach (panel, Props, or Object menu)
Delete definition	× on the symbol row (instances may draw empty if master is gone)

Creating a symbol replaces the selection with an instance of the new master.

6.5 Plug (Plugins)

Lists registered plugins and runs their commands. Ships with three built-ins (see [Plugins](#)).

6.6 Align & Boolean (sidebar)

Always visible in Edit mode under the right panels.

Align (two or more objects)

Button	Align
L	Left
C	Horizontal center
R	Right
T	Top
M	Vertical middle
B	Bottom

Distribute-horizontal / distribute-vertical exist in code but are not exposed in this UI build.

Boolean

Button	Operation
Unite	Union
Inter	Intersection
Sub	Subtract (first minus others)

Xor	Exclusive or / exclude
-----	------------------------

Live preview: hover (or focus) a boolean button to see a translucent lime ghost of the result. Click to commit. Preview clears when the pointer leaves the boolean group. The same preview works from **Object** menu items.

Booleans require at least two selected **filled** shapes that can be flattened (rect, ellipse, closed path, groups of those). Results become polygonal path geometry (curves are sampled).

7. Menus

7.1 File

Command	Behavior
Open...	Opens .savage or .svg into Edit. Raster types switch to Convert mode (load the image via the Convert drop zone / Open Image).
Save Project...	Writes full document JSON as .savage
Export SVG...	Writes serialized SVG
Export PNG...	Renders via Rust (resvg), typically at 2× scale

7.2 Edit

Undo, Redo, Copy, Paste — see shortcuts table.

7.3 Object

Command	Behavior
Group / Ungroup	Group selection; ungroup primary selection
Unite / Intersect / Subtract / Exclude	Boolean ops + live hover preview
Simplify Path	Reduces complexity of selected paths
Convert Text to Outlines	Contour extraction from text

Make Clip Mask	Last selected shape is the mask; earlier selection is clipped; mask is hidden
Release Clip Mask	Restores mask visibility; clears clip links on selection
Create Symbol / Detach Symbol	Symbol workflows
Commit Shape Builder	Finalize Shape Builder session
New Artboard	Adds a board and fits the view

Clip mask recipe: select content first, then the mask shape last → **Make Clip Mask**. Mask must be a path, rect, or ellipse.

7.4 View

Fit Artboard, Fit All Artboards, Fit Selection, Zoom 50% / 100% / 200%.

7.5 Help

Command	Behavior
User Manual (PDF)...	Opens the bundled <code>USER_MANUAL.pdf</code> in the system PDF viewer

Regenerate the PDF after editing the Markdown with `pnpm manual:pdf`.

8. Advanced editing

8.1 Artboards (multi-board documents)

Treat artboards as design frames (icons set, poster variants, mobile screens). Export SVG can include all boards or focus workflows around the active board (fit / PNG export uses the document serialization path — prefer saving `.savage` when board metadata matters).

Activate a board before drawing if you want new content mentally scoped to that frame; nodes are free-positioned in world space (artboards are guides/frames, not strict clip parents unless you clip).

8.2 Symbols / components

When to use: repeating logos, icons, UI chrome.

Instance vs detach:

- Keep instances linked when you want many copies of one master definition in the file.
- **Detach** expands an instance into ordinary editable nodes (independent copies).

SVG export emits `<symbol>` definitions and `<use>` instances where possible.

8.3 Shape Builder (professional use)

Ideal for logo marks and icon negative space:

1. Build overlapping primitives.
2. Run Shape Builder; discard outer waste; keep the silhouette regions.
3. Commit once.
4. Optionally Simplify Path and convert remaining pieces into a Symbol.

Region decomposition uses boolean intersect/subtract over selection subsets (capped at four shapes). Complex illustrations may need staged builds (boolean first, then Shape Builder).

8.4 Mesh gradients

1. Select a filled shape.
2. Props → Fill → **Mesh**.
3. Edit corner / control-point color swatches.

Canvas preview bilinear-samples the mesh into a pattern clipped to the shape.

Export note: SVG mesh gradients are not a portable SVG standard feature. SaVaGe approximates with a `<pattern>` of polygons. For maximum fidelity across editors, keep a `.savage` master and export PNG for final raster deliverables when mesh appearance is critical.

8.5 Variable-width strokes

Uniform strokes use `StrokeStyle.width`. Paths may also store per-point `strokeWidth`.

Method	Result
Props → Taper ends	Automatic profile
Direct Select + Alt-drag	Manual width at a point
Props → Clear profile	Back to uniform

When any point width differs from the base width, SaVaGe draws a **filled ribbon** outline instead of a canvas `stroke()`. That ribbon is editable geometry only insofar as the centerline points and widths change — it is not a separate expanded outline node unless you otherwise convert it.

8.6 Perspective grid

Toolbar **Persp**:

Mode	Guides
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Off	No perspective overlay
1-pt	Horizon + fan from one vanishing point
2-pt	Horizon + fans from two vanishing points

Enable **Snap** to pull drawing coordinates onto the nearest guide ray (threshold scales with zoom).

Important: Snap here means **perspective ray snap**, not general document-grid snap. Grid is a visual overlay toggled separately.

Vanishing points are initialized from the document view box; v0.1 does not expose interactive VP handles in the UI.

8.7 Brushes compared

Brush	Geometry	Typical use
Calligraphy (B)	One closed ribbon path	Expressive strokes, ink feel
Pattern	Group of oriented motifs	Borders, decorative paths
Scatter	Group of jittered motifs	Foliage, texture accents

9. Files, export & clipboard

9.1 Formats

Format	Role	Fidelity
.savage	Native project (JSON scene graph)	Full: artboards, symbols, mesh, effects, widths
.svg	Interchange / web	High for solids and basic gradients; mesh approximated; some import gaps
PNG	Raster export	Visual snapshot (scaled render)
Raster inputs	Convert only	Become vectors after trace

9.2 Recommended save strategy

1. Always keep a **.savage** working file.
2. Export SVG for handoff to web / other editors.

3. Export PNG for previews, social, or mesh-critical stills.
4. Re-open SVG when you must — but expect gradient/ `url(#...)` paints to simplify on import (solids preferred for SVG round-trips in v0.1).

9.3 Clipboard

Copy writes:

- SVG text to the system clipboard (when permitted)
- An internal session snapshot (including symbol masters referenced by copied instances)

Paste prefers the internal snapshot, then falls back to parsing SVG from the clipboard. Pasted objects offset by +24,+24 world units.

9.4 Duplicate

Ctrl+D duplicates the selection in-document (fast iteration without clipboard).

10. Plugins

10.1 Built-in commands (Plug tab)

Plugin	What it does
Duplicate & Offset	Copies selected non-group objects +24,+24
Randomize Fills	Assigns random palette fills to selected fillable nodes
Add Guide Rect	Inserts a translucent cyan rectangle as a layout guide

Commands toast status messages when finished.

10.2 For developers (plugin API)

SaVaGe exposes a lightweight in-app plugin registry:

- `registerPlugin({ id, name, description?, commands[] })`
- Each command receives an API: selection, get/update/add/delete nodes, export SVG string, notify, markDirty
- `runPluginCommand(pluginId, commandId)`
- UI lists whatever is registered (built-ins register on app start)

This is an in-process TypeScript extension surface (not a sandboxed third-party store). Suitable for studio scripts and experimental tools.

11. Keyboard shortcuts cheat sheet

Global (canvas focused)

Shortcut	Action
V	Select
A	Direct Select
H	Pan
R	Rectangle
O	Ellipse
L	Line
P	Pen
B	Calligraphy Brush
S	Shape Builder
T	Text
Space (hold)	Temporary pan
Ctrl+Z	Undo
Ctrl+Shift+Z / Ctrl+Y	Redo
Ctrl+A	Select all
Ctrl+D	Duplicate
Ctrl+C / Ctrl+V	Copy / Paste
Ctrl+0	Fit artboard
Ctrl+1	Zoom 100%
Ctrl+2	Fit selection
Delete / Backspace	Delete selection
Arrows	Nudge 1 px
Shift+Arrows	Nudge 10 px

Tool-specific

Shortcut	Context	Action
Enter / Esc	Pen	Finish path
Backspace	Pen	Drop last point
Enter	Shape Builder	Commit
Esc	Shape Builder	Cancel
Enter / Esc	Text overlay	Commit / cancel
Shift	Select / shapes	Multi-select / constrain
Alt	Shape tools	Draw from center
Alt+drag	Direct Select	Edit point stroke width

Menu items do not display accelerator labels in the title bar; shortcuts are handled by the editor viewport.

12. Known limitations & best practices

12.1 Limitations (v0.1)

1. **Mesh** → **SVG** is an approximation (`<pattern>` polygons), not a native SVG mesh.
2. **Opening SVG** currently restores solid/ `none` paints reliably; `url(#gradient)` fills may not fully round-trip — use `.savage` for gradient/mesh masters.
3. **File** → **Open** on a raster switches to Convert but does not auto-attach the file to the drop zone; use Convert's Open Image / drag-drop.
4. **Snap** is perspective-ray snap only (when Persp ≠ Off).
5. **Props W/H** are read-only; transform on canvas.
6. **Shape Builder** supports up to four selected flattenable shapes.
7. **Text** → **outlines** uses raster contouring, not TrueType/CFF glyph extraction.
8. **Help** menu is empty; distribute-align UI is not exposed yet.
9. Pattern/Scatter brush parameters are fixed in this release.

12.2 Best practices

Goal	Practice
Logo from PNG	Logo preset → Simplify → Boolean/Shape Builder → Symbol
Editable master file	Always save <code>.savage</code>
Web SVG	Prefer solids + linear/radial; expand critical meshes to PNG or simplified geometry

Clean booleans	Work with closed fills; convert text/strokes to fills first
Performance	Watch status-bar ms; simplify dense converts; hide unused boards' complexity by deleting unused paths
Repeat UI chrome	Symbols over raw duplicates
Perspective sketching	Persp 1-pt/2-pt + Snap on while blocking shapes

12.3 Troubleshooting

Symptom	Try
Boolean fails	Ensure ≥ 2 filled closed shapes; ungroup if needed; convert strokes to filled ribbons or shapes
Shape Builder empty	Select overlapping flattenable shapes first; max 4
Mesh looks banded in SVG	Expected approximation; use PNG or .savage
Paste did nothing	Focus canvas; ensure clipboard has SVG or a prior SaVaGe copy
Heavy lag	Simplify paths; reduce convert color precision; hide effects temporarily

13. Glossary

Term	Meaning in SaVaGe
Artboard	Named frame/guide with size and background; multi-board documents supported
Boolean	Path union / intersection / difference / xor via Rust <code>i_overlay</code>
Clip mask	One shape clipping others via <code>clipPath</code>
Instance	Placed reference to a symbol master
Mesh	Grid of colored control points interpolated as a fill
Ribbon	Variable-width stroke drawn as a filled outline around a centerline
Scene graph	Typed document nodes (<code>SvgDocument</code>) — source of truth while editing
Shape Builder	Interactive keep/discard of boolean regions
Speckle	Tiny unwanted regions removed during tracing

vtracer	Rust vectorization engine behind Convert
.savage	Native JSON project format

Appendix A — Suggested learning path

1. Convert a simple logo PNG.
 2. Recolor with Props; rename layers.
 3. Draw primitives; Unite; Shape Builder.
 4. Create a Symbol; place twice; detach one.
 5. Try Brush, Pattern, Scatter.
 6. Enable 1-pt perspective and sketch with Snap.
 7. Save **.savage** ; export SVG and PNG; reopen both and compare fidelity.
 8. Run Plug → Randomize Fills once for fun, then Undo.
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Appendix B — Support & product identity

SaVaGe blends **SVG** with a bold “savage” craft aesthetic: graphite UI (**#0B0D10**), lime accent (**#B8FF3C**), and expressive typography (Syne / DM Sans).

For development builds, issues, and roadmap context, see the project plan (**savage_svg_studio_2bf62e1a.plan.md**) and **checklist.md** in the repository.

End of user manual.